

Cardiovascular Disease Prediction

Group 15



PROBLEM



Cardiovascular diseases are a major cause of death worldwide. With more people at risk due to unhealthy lifestyles and genetics, it's important to find ways to identify those at high risk. This project focuses on using data mining techniques to analyze a dataset related to cardiovascular diseases. The goal is to classify individuals based on their risk factors, helping to predict who may develop CVDs.

DATASET

- Dataset Overview:

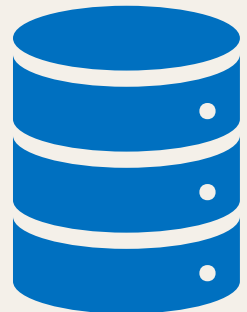
Number of Objects: 70000

Number of Attributes: 14.

- Attribute Details:

Data types: Integer, Decimal data

Missing Values: No missing values



DATA VISUALIZATION

Purpose of Data Visualization in Data Mining:

- **Understanding Patterns and Trends**

Uncovering hidden patterns, relationships, and trends in the dataset.

- **Simplifying Complex Data**

Transforming raw data into a visual format.

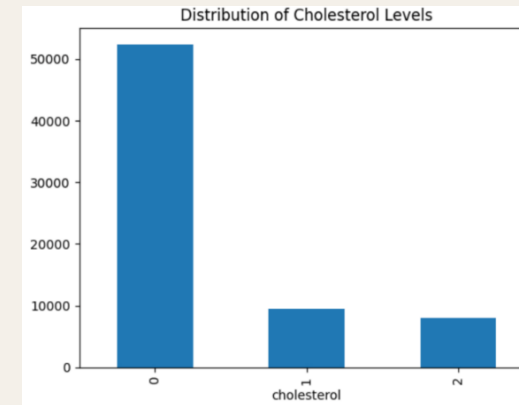
- **Supporting Decision-Making**

Provides insights that enable informed decisions in various fields.

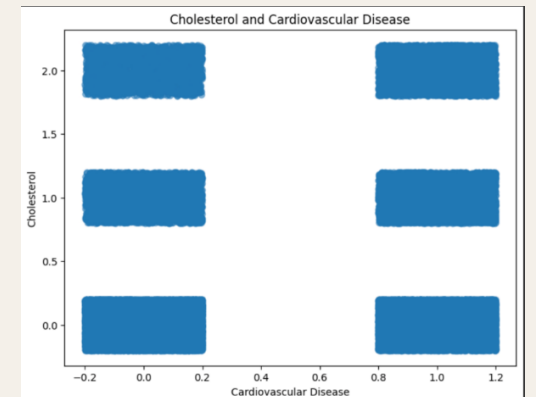
- **Communicating Results**

Enhances the presentation of findings clearly and compellingly for diverse audiences.

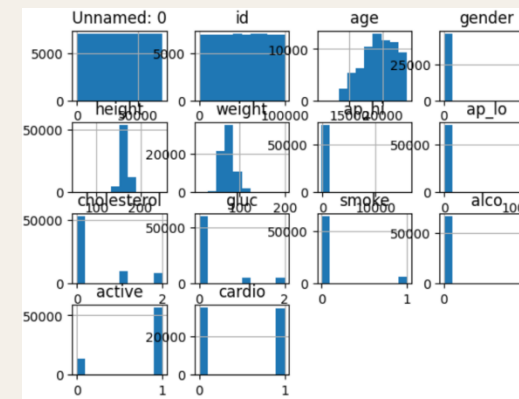
Bar Plots



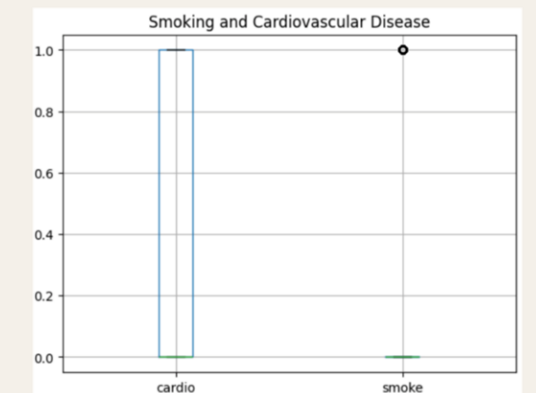
Scatter Plots



Histogram



Boxplots



DATA MINING TECHNIQUES

Classification

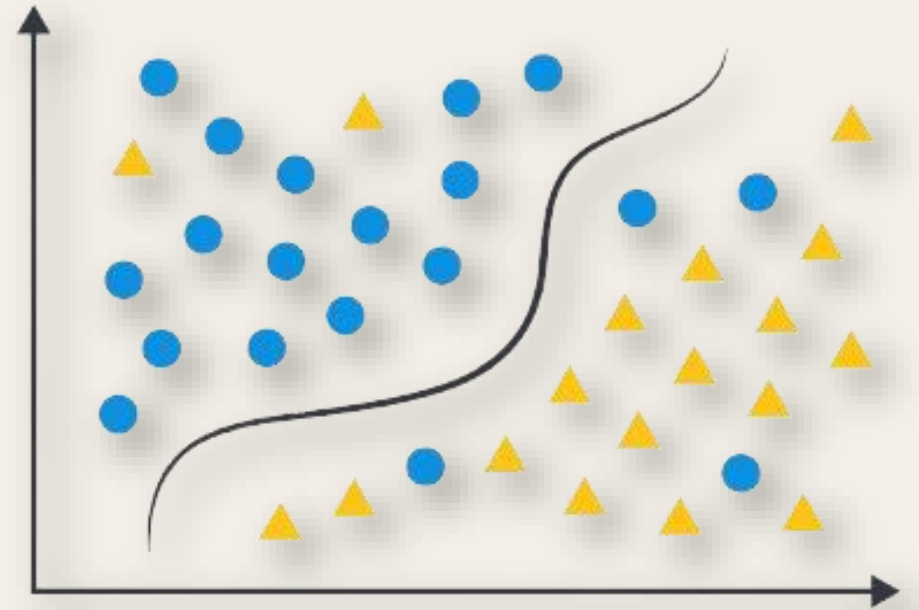


Clustering



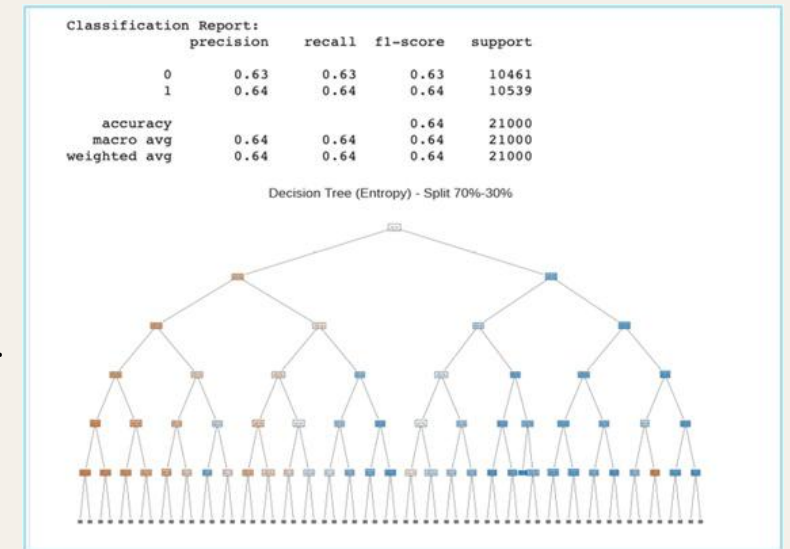
CLASSIFICATION

- **Supervised Learning.**
- **Selection Measures Used:** Gini Index and Entropy.
- **Data Splits Tested:** 90%-10%, 80%-20%, and 70%-30%.



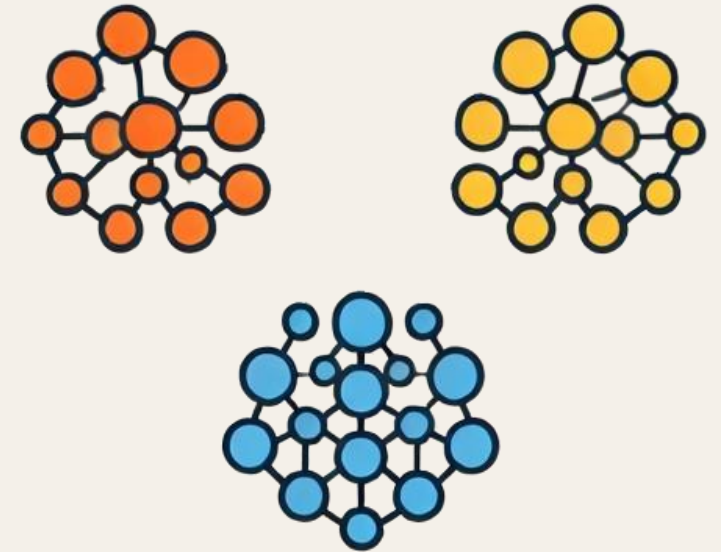
CLASSIFICATION FINDINGS

- Higher training data ratio improved precision, recall, and F1-score.
- The **best accuracy** was achieved with a data split of **70% training data and 30% test data**.
- Confusion Matrix showed clear accuracy in predicting target classes.
- Gini criterion showed balanced splits.
- Entropy criterion focused more on reducing uncertainty in final nodes.



CLUSTERING

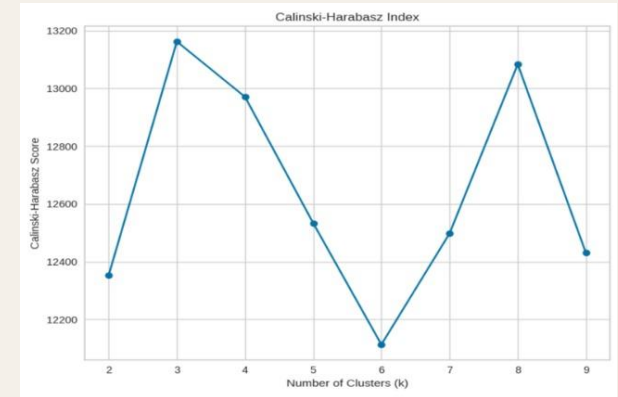
- **Unsupervised Learning.**
- **Method Used:** K-Means clustering.
- **Evaluation:** Silhouette width for $K=2,4,8$.



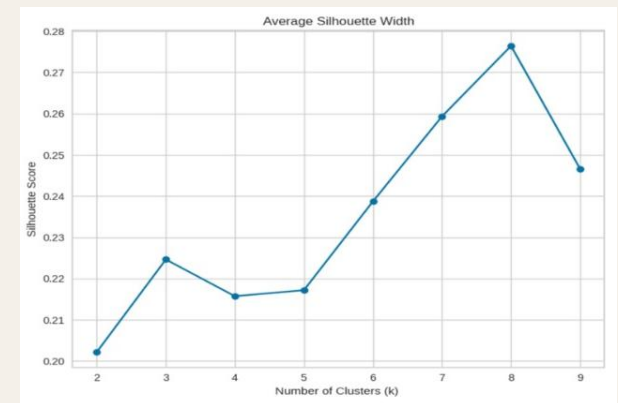
CLUSTERING FINDINGS

- K-Means Clustering was evaluated to determine the optimal number of clusters.
- **Silhouette Analysis:** Suggested $k=8$ for balanced cluster quality.
- The Calinski-Harabasz Index showed distinct clusters, peaking at $k=3$ and second highest at $k=8$.
- **Elbow Method:** Optimal clusters identified at $k=8$ based on reduced improvement in WSS.

Calinski-Harabasz Index



Silhouette Analysis



THANK YOU!